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1. Introduction

The hotel industry faces persistent challenges including high booking cancellation rates, variable Average Daily Rates (ADR), and diverse customer behaviors. This assignment applies machine learning to the Hotel Booking Demand dataset (Mostipak, 2020) to generate actionable insights for reducing revenue loss, optimizing pricing, and improving customer segmentation. Following the CRISP-DM methodology (Chapman et al., 2000), three tasks were completed: classification to predict cancellations, regression to forecast ADR and total revenue, and clustering to identify customer segments. This supplementary report summarizes the methodology, key findings, model comparisons, and business recommendations. Full implementation details, code, and visualizations are provided in the accompanying Jupyter Notebooks.

2. Dataset Overview and Preprocessing

2.1 Dataset

The Hotel Booking Demand dataset (Mostipak, 2020) contains 119,390 hotel booking records from a city hotel and a resort hotel in Portugal, covering arrivals between July 2015 and August 2017. Each record has 32 features spanning booking metadata, guest characteristics, room details, and cancellation status.

2.2 Key Data Quality Challenges

Three issues required careful handling before modelling could begin:

- Missing values: 'company' had 94.3% missing entries and was dropped. 'agent' (16.3% missing) was filled with 0, representing direct bookings. 'children' (0.003% missing) was filled with 0. 'country' (0.4% missing) was filled with 'Unknown' to preserve row count.
- Data leakage: 'reservation status' and 'reservation status date' are directly derived from the target variable (is_canceled) and would cause unrealistically high model performance if included. Both were dropped from all three tasks.
- Anomalous rows: Bookings with zero guests (adults=0, children=0, babies=0) and negative ADR values were removed as they represent data entry errors (roughly 180 rows total; negligible loss).

2.3 Feature Engineering

Seven features were engineered from raw columns to capture business-relevant signals:

Total_guests, total_nights, weekend_ratio, season, lead_time_category, room_mismatch, total_revenue.

Each notebook contains task-specific preprocessing.

- Classification: used SMOTE oversampling.
- Regression: capped ADR outliers at IQR upper fence

- Clustering: retained full dataset without train/test/split, since unsupervised learning has no labels to hold.

3. Task Summaries & Model Performance

Task 2: Classification - Predicting Cancellations

The binary target *is_canceled* showed a 37% positive class rate. Data preparation involved handling missing values, feature engineering (total_guests, total_nights, seasonal features), StandardScaler normalization, and SMOTE for class imbalance (Chawla et al., 2002). Three algorithms were trained and tuned using GridSearchCV: Logistic Regression, Random Forest, and XGBoost.

XGBoost achieved the strongest performance on the test set (F1-score = 0.86, ROC-AUC highest). An optimized decision threshold of 0.62 improved recall to ~85%, balancing the business cost of missed cancellations versus unnecessary retention offers. SHAP values (Lundberg and Lee, 2017) highlighted lead time, non-refundable deposit type, previous cancellations, and absence of special requests as key drivers. Cross-validation confirmed model stability.

Task 3: Regression - Predicting ADR and Revenue

ADR was the primary target (highly right-skewed with outliers). Preprocessing included IQR capping, scaling, and log-transform experiments. Models evaluated were Linear Regression, Ridge, Lasso, Random Forest, and XGBoost.

XGBoost outperformed others ($R^2 = 0.83$, lowest RMSE). Tree-based ensembles effectively captured interactions such as season x room type x hotel type, which linear models could not. SHAP analysis (Lundberg and Lee, 2017) and quantile regression for prediction intervals enhanced interpretability and practical utility. A secondary *total_revenue* model was successfully developed using the same pipeline.

Task 4: Clustering - Customer Segmentation

Features included *lead_time*, *total_guests*, *total_nights*, *ADR*, *market_segment*, *hotel*, and *meal*. After scaling and PCA (retaining ~90% variance), K-Means, Agglomerative Hierarchical (Ward linkage), and DBSCAN were compared using Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index, and Gap Statistic.

K-Means (K=4) performed best, producing compact and well-separated clusters. DBSCAN generated excessive noise due to the dataset's relatively uniform density in feature space. Four interpretable customer personas emerged: Last-Minute City Explorers, Premium Family Planners, Corporate Road Warriors, and Group Coordinators.

4. Model Comparison and Fine-Tuning Reflections

Across supervised tasks, XGBoost consistently delivered superior results due to its ability to model complex non-linear interactions and handle mixed data types effectively. Hyperparameter tuning via GridSearchCV (classification) and RandomizedSearchCV (regression) was essential. In classification, business-cost-aware threshold optimization (moving from default 0.5 to 0.62) significantly improved practical utility.

Linear models served as strong baselines and offered interpretability but underperformed due to violated assumptions (additivity in regression, linear separability in classification). In clustering, multiple internal validation metrics were necessary because of the absence of ground-truth labels. PCA proved valuable by reducing noise from one-hot encoded features, improving Silhouette scores by approximately 15-20% compared to clustering on the full scaled feature set.

Key Challenges Addressed:

- Class imbalance and skewness were mitigated using SMOTE (Chawla et al., 2002) and targeted transformations.
- Stability was verified through cross-validation and Adjusted Rand Index across random seeds.
- Overfitting risks were managed via regularization, early stopping (XGBoost), and careful feature selection to prevent leakage.

Limitations include the absence of external variables (competitor pricing, local events, real-time occupancy) and potential temporal effects. Future work could incorporate time-series modelling and real-time deployment.

5. Business Recommendations

Key Findings identified were:

Finding	Business Implication
Long lead time (>90 days) is the strongest cancellation predictor	Early bookings carry significantly higher cancellation risk
Non-refundable deposits show high cancellation rates	Current policy is not effective at preventing cancellations
Previous cancellations strongly predict future cancellations	Flag repeat cancellers immediately
Zero special requests increase cancellation likelihood	Guests with no requests are less committed
Repeated guests have the lowest cancellation rate	Loyalty programs significantly reduce risk
Summer season & Resort Hotel drive the highest ADR	Focus premium pricing and promotions here
Premium room types (H, G, F) command much higher ADR	Prioritize upselling at check-in
Online TA segment has both high volume and high cancellation rate	Strengthen deposit policy or incentivize direct bookings
Four distinct customer segments identified	Each segment requires tailored pricing, marketing, and retention strategies

The three models together provide a powerful decision-support framework:

1. Proactive Cancellation Management

Integrate the tuned XGBoost classifier into the booking system. Flag high-risk bookings (probability > 0.62) for targeted retention offers. Prioritize interventions for the "Last-Minute City Explorer" segment, which shows elevated cancellation risk. Projected annual revenue protection exceeds £13 million based on conservative assumptions scaled from the dataset.

2. Dynamic Pricing and Revenue Optimization

Deploy the XGBoost regressor for real-time ADR guidance. Use prediction intervals for conservative budgeting and aggressive overbooking strategies. Differentiate pricing by cluster: premium rates and minimum-stay requirements for "Premium Family Planners" (high ADR, long lead time, Resort Hotel), while offering controlled last-minute deals for price-sensitive segments.

3. Segment-Specific Marketing and Loyalty

Develop tailored strategies for each persona:

- **Premium Family Planners:** Early-bird packages, family upgrades, and loyalty rewards.
- **Corporate Road Warriors:** Corporate agreements, express services, and weeknight incentives.
- **Last-Minute City Explorers:** OTA partnerships with stricter cancellation policies.

Combining cluster labels with cancellation predictions enables highly personalized and cost-effective interventions.

4. Operational Integration and Monitoring

Assign cluster labels and risk scores in real time to every new booking. Retrain models quarterly. This integrated system supports better staffing forecasts, overbooking decisions, and guest experience personalization, ultimately improving both revenue and satisfaction.

6. Conclusion

This assignment successfully applied **classification, regression, and clustering** techniques to deliver actionable insights for hotel operations. XGBoost emerged as the strongest supervised model, while K-Means provided robust, interpretable customer segments. The combination of SMOTE for imbalance handling (Chawla et al., 2002) and SHAP for model interpretability (Lundberg and Lee, 2017) strengthened both technical rigor and business relevance. By addressing cancellations, pricing, and segmentation in an integrated manner, the developed solutions directly support the hotel's goals of reducing revenue loss, optimizing pricing strategies, and enhancing targeted marketing. Implementation of these models has the potential to deliver substantial financial benefits and a more personalized guest experience.

7. References

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